

El Salvador Competition Superintendency

In some countries, computational antitrust is making its way into the future for the competition agencies, although the use of computational tools⁴ as a proactive response for detecting anti-competitive practices a modern is not yet implemented. More experienced competition agencies are just beginning to acquire the technical expertise to develop and use them.⁵ Younger agencies rely on technological advances for their reactive response, i.e., dawn raids for investigative procedures.

The Competition Superintendency (Hereinafter SC for its acronym in Spanish) from El Salvador has been active since 2006, and even though it has budget limitations, is always finding new ways to improve and modernize its investigative tools. Its advances in computational antitrust are small compared to other jurisdictions, yet still very significant. They relate more to identifying possible anticompetitive practices through traditional ways and using computer forensics than to the use of algorithms to detect illegal practices for now.

Since 2020, the SC has been developing a market intelligence tool that collects information with the intent of guiding and promoting the focused initiation of effective informal investigations, aimed at detecting and punishing cartels through proactive and reactive investigation methods. This proactive tool consists of 4 components: public procurement; markets in general and collecting relevant public information; the use of ICTs for market monitoring; and reactive mechanisms. The tool still uses traditional ways of obtaining and analyzing information, but the aim is to elevate it to a more complex tool, that integrates both traditional and modern tools that can be used in all markets, including the digital ones. Since this tool is still in development, the SC has focused its limited resources to reinforce the reactive tools for combating cartels and other anti-competitive practices.

At the end of 2021 and the beginning of 2022, the CS acquired specific hardware and software to collect, process, and analyze digital evidence, therefore improving data gathering on dawn raids and its following analysis. This equipment, although it may not be considered a proper computational antitrust tool according to theory, is a giant step forward regarding the normal use of technological tools for third-world competition agencies, such as CS.

⁴ Defined by Dr. Thibault Schrepel as computer-based problem-solving methods, Thibault Schrepel, *A First Look at Computational Antitrust* (12 September 2021), <https://perma.cc/39HZ-CLYN>.

⁵ Thibault Schrepel, *Computational Antitrust: An Introduction and Research Agenda*, 1 STANFORD COMP. ANTITRUST (2021).

Among the hardware acquired are a TXi Forensic Imager,⁶ which allows reducing the time of making forensic images; an UltraBlock Kit which permits read-only access to data storage devices without compromising the integrity of the data;⁷ a Star Tech Drive Eraser,⁸ that allows the reuse of the storage devices used by the CS in one case, to make them available for future cases; a Forensic Recovery of Evidence Device (FRED),⁹ a workstation that allows connecting the storage media (SATA hard drives, SAS hard drive, USB and CD/DVD/Blu-ray) that contains the digital evidence collected by the SC and includes blockers of writing to guarantee the protection and integrity of the files subject to analysis and thereby preserve the chain of custody of said evidence.

Regarding software, the CS now has an FTK Imager,¹⁰ which is a free tool that allows forensic imaging of computer equipment saves an image of a hard disk in one file or in segments that may be reconstructed later, and helps analyze the digital evidence collected in the raid. The CS also uses Autopsy 4.19.3, an open-source software that facilitates the analysis of digital evidence and allows the identification, selection, and organization of the elements of main interest for the investigation.¹¹ Some of this equipment was used for the first time in a recent dawn raid made by the CS and has already proven useful for facilitating the search for keywords in the information gathered. It is expected that these tools will improve the CS's ability to obtain and analyze data from companies that are being investigated for anti-competitive practices.

Although the proper use of computational tools like algorithms and artificial intelligence seem far away from the reach of small-countries' competition agencies like the CS, the authority is always looking forward to learning about the recent development in this Antitrust 3.0 era, hoping that the advances in technology and the knowledge that initiatives like the Computational Antitrust project provides may help to find ways for these tools to get closer to grasp for the CS, so that it can better fulfill its purpose of protecting, promoting and guaranteeing competition, as required by Salvadorian competition law.

⁶ A forensic imaging solution that acquires data faster and from more media types, without sacrificing ease-of-use or portability. opentext Security, <https://perma.cc/DU6E-J9JQ>.

⁷ It is a hardware-based write blocker that allows a USB mass storage device (i.e., flash drives, thumb drives, external USB hard drives, etc.) to be connected to a personal computer for forensic image acquisition and analysis and can guarantee the protection of the data chain of custody. Provides outstanding imaging speed. Digital Intelligence, <https://perma.cc/6QC8-KFFA>.

⁸ This standalone eraser dock erases drives without connecting to a host computer, making it easy to re-use or re-purpose the drive. StarTech.com, <https://perma.cc/LHW4-4C37>.

⁹ FRED systems are high-performance, forensic acquisition, analysis, and processing platforms that set the standard for forensic acquisition and analysis workstations. Digital Intelligence, <https://perma.cc/9HMT-CJHT>.

¹⁰ Quickly assess electronic evidence by obtaining forensic images of computer data, without making changes to the original evidence. AccessData, <https://perma.cc/DAV3-Q8FB>.

¹¹ Autopsy® is a premier end-to-end open-source digital forensics platform. Autopsy, <https://perma.cc/T7MP-FGDB>.